

Strengthening System Security

Tools Used

- Kali Linux
- PAM / libpam-pwquality
- auditd / auditctl / ausearch
- sudo / systemctl
- chage / passwd
- grep / awk / find / getent

Exercise 1: Locate and Open the adduser.conf File

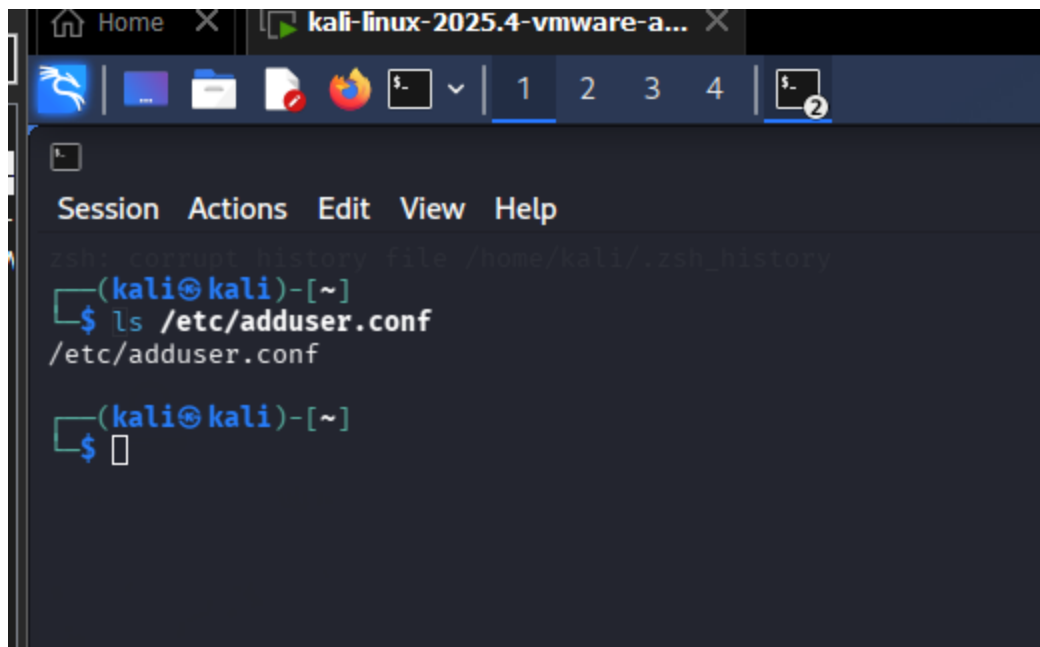
Command Used:

```
ls /etc/adduser.conf
```

Opened File Editor

```
nano
```

```
sudo nano /etc/adduser.conf
```



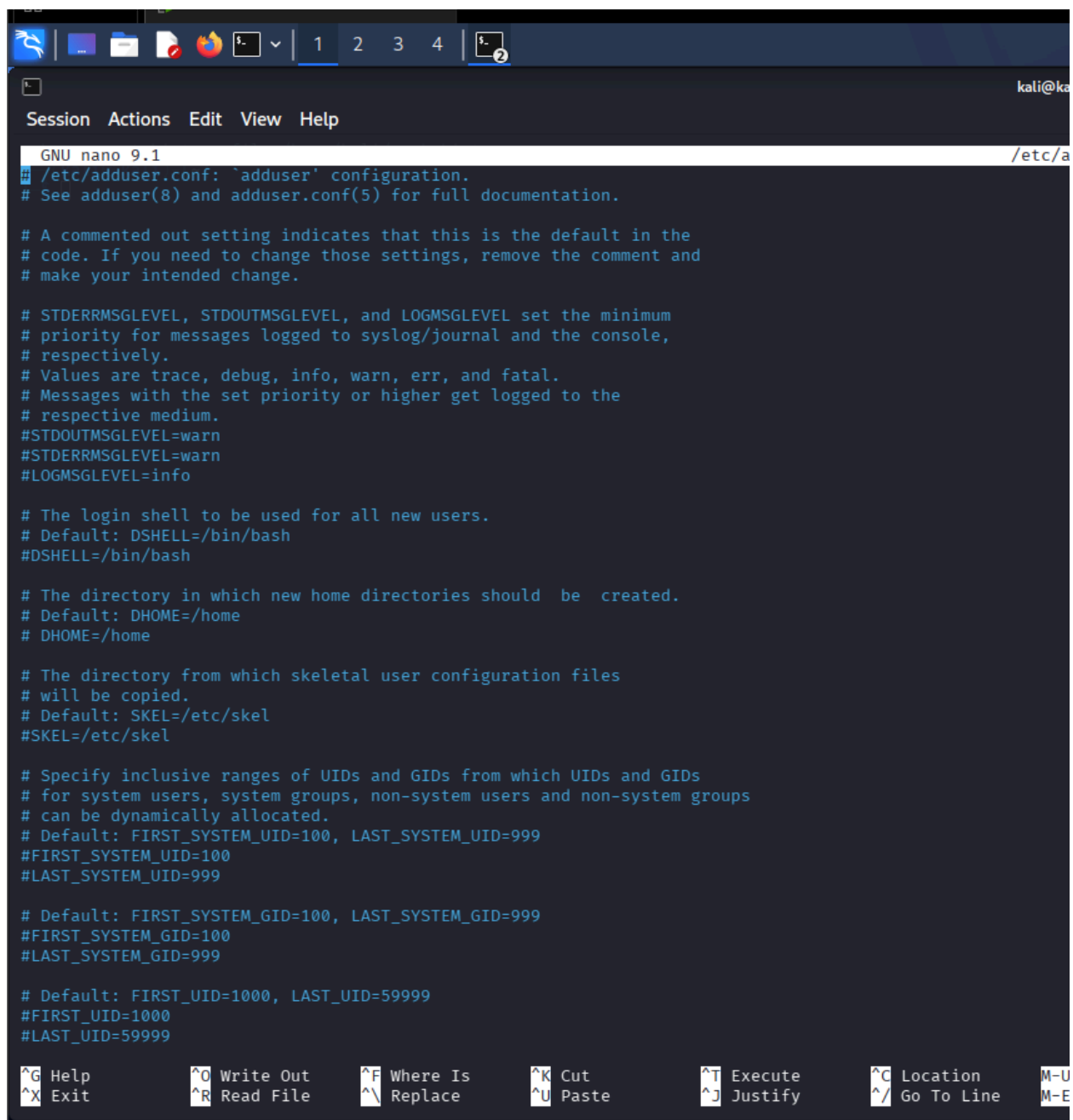
A terminal window titled "kali-linux-2025.4-vmware-a..." with a dark theme. The terminal shows the following commands and output:

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ ls /etc/adduser.conf
/etc/adduser.conf
(kali@kali)-[~]
$
```



A terminal window showing the following commands and output:

```
(kali@kali)-[~]
$ sudo nano /etc/adduser.conf
[sudo] password for kali:
(kali@kali)-[~]
$
```



```
GNU nano 9.1 /etc/a
# /etc/adduser.conf: 'adduser' configuration.
# See adduser(8) and adduser.conf(5) for full documentation.

# A commented out setting indicates that this is the default in the
# code. If you need to change those settings, remove the comment and
# make your intended change.

# STDERRMSGLEVEL, STDOUTMSGLEVEL, and LOGMSGLEVEL set the minimum
# priority for messages logged to syslog/journal and the console,
# respectively.
# Values are trace, debug, info, warn, err, and fatal.
# Messages with the set priority or higher get logged to the
# respective medium.
#STDOUTMSGLEVEL=warn
#STDERRMSGLEVEL=warn
#LOGMSGLEVEL=info

# The login shell to be used for all new users.
# Default: DSHELL=/bin/bash
#DSHELL=/bin/bash

# The directory in which new home directories should be created.
# Default: DHOME=/home
# DHOME=/home

# The directory from which skeletal user configuration files
# will be copied.
# Default: SKEL=/etc/skel
#SKEL=/etc/skel

# Specify inclusive ranges of UIDs and GIDs from which UIDs and GIDs
# for system users, system groups, non-system users and non-system groups
# can be dynamically allocated.
# Default: FIRST_SYSTEM_UID=100, LAST_SYSTEM_UID=999
#FIRST_SYSTEM_UID=100
#LAST_SYSTEM_UID=999

# Default: FIRST_SYSTEM_GID=100, LAST_SYSTEM_GID=999
#FIRST_SYSTEM_GID=100
#LAST_SYSTEM_GID=999

# Default: FIRST_UID=1000, LAST_UID=59999
#FIRST_UID=1000
#LAST_UID=59999

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E
```

The adduser.conf file was successfully located and opened for review. This provided an overview of the default configuration settings used for creating new user accounts in Linux.

Exercise 2: Analyze Key Configuration Parameters

Commands Used to search:

1. Default Group Settings:
GROUP=100.
2. Home Directory Settings:
DHOME=/home

Skel Directory:

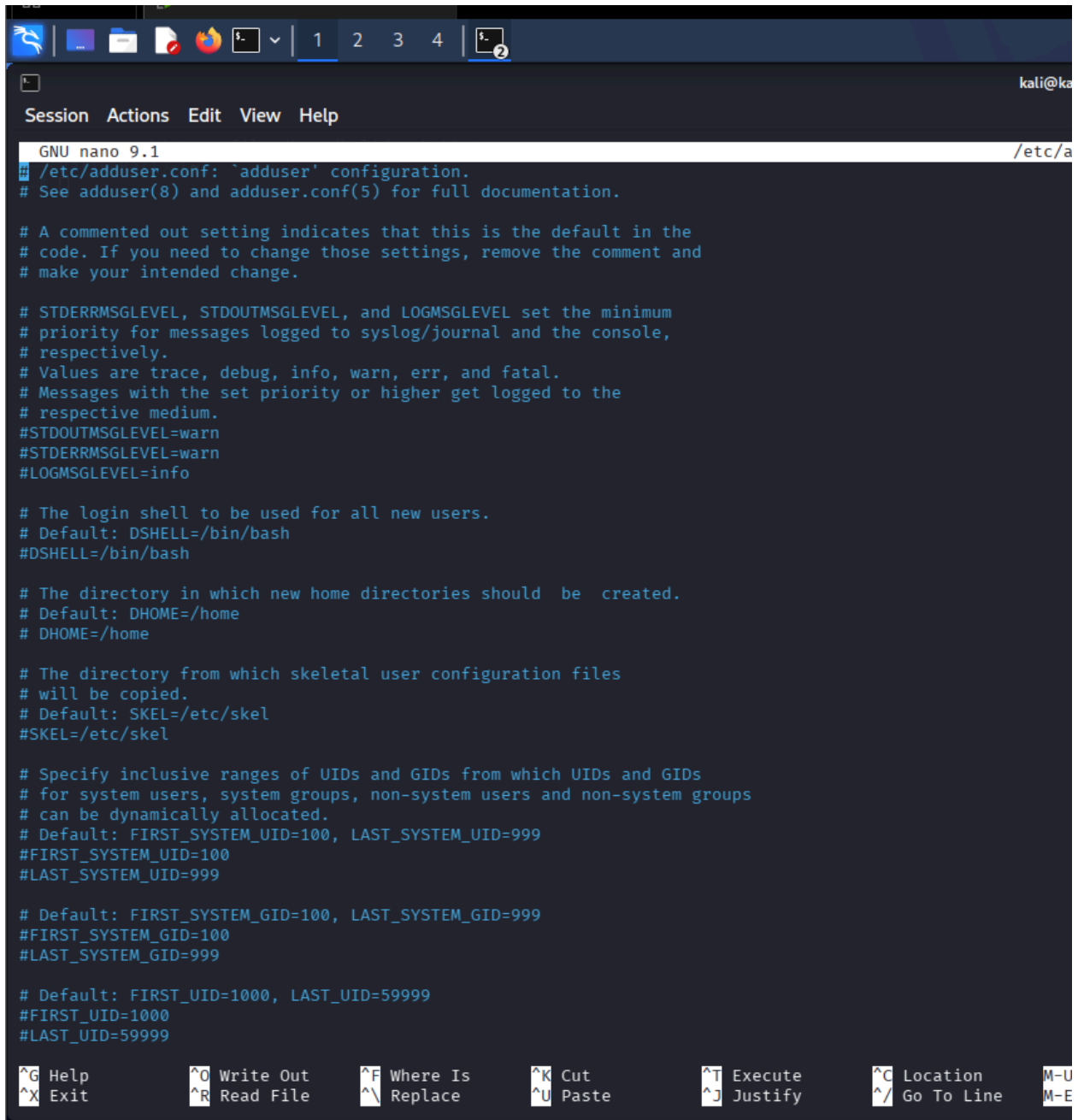
- Locate

SKEL=/etc/skel

4. Default Shell:
DSHELL=/bin/

5. Useful Fields in adduser.conf:

- FIRST_UID
- LAST_UID
- FIRST_GID
- LAST_GID
- USERGROUPS
- HOME
- LOGIN_SHELL



```
GNU nano 9.1 /etc/a
# /etc/adduser.conf: 'adduser' configuration.
# See adduser(8) and adduser.conf(5) for full documentation.

# A commented out setting indicates that this is the default in the
# code. If you need to change those settings, remove the comment and
# make your intended change.

# STDERRMSGLEVEL, STDOUTMSGLEVEL, and LOGMSGLEVEL set the minimum
# priority for messages logged to syslog/journal and the console,
# respectively.
# Values are trace, debug, info, warn, err, and fatal.
# Messages with the set priority or higher get logged to the
# respective medium.
#STDOUTMSGLEVEL=warn
#STDERRMSGLEVEL=warn
#LOGMSGLEVEL=info

# The login shell to be used for all new users.
# Default: DSHELL=/bin/bash
#DSHELL=/bin/bash

# The directory in which new home directories should be created.
# Default: DHOME=/home
# DHOME=/home

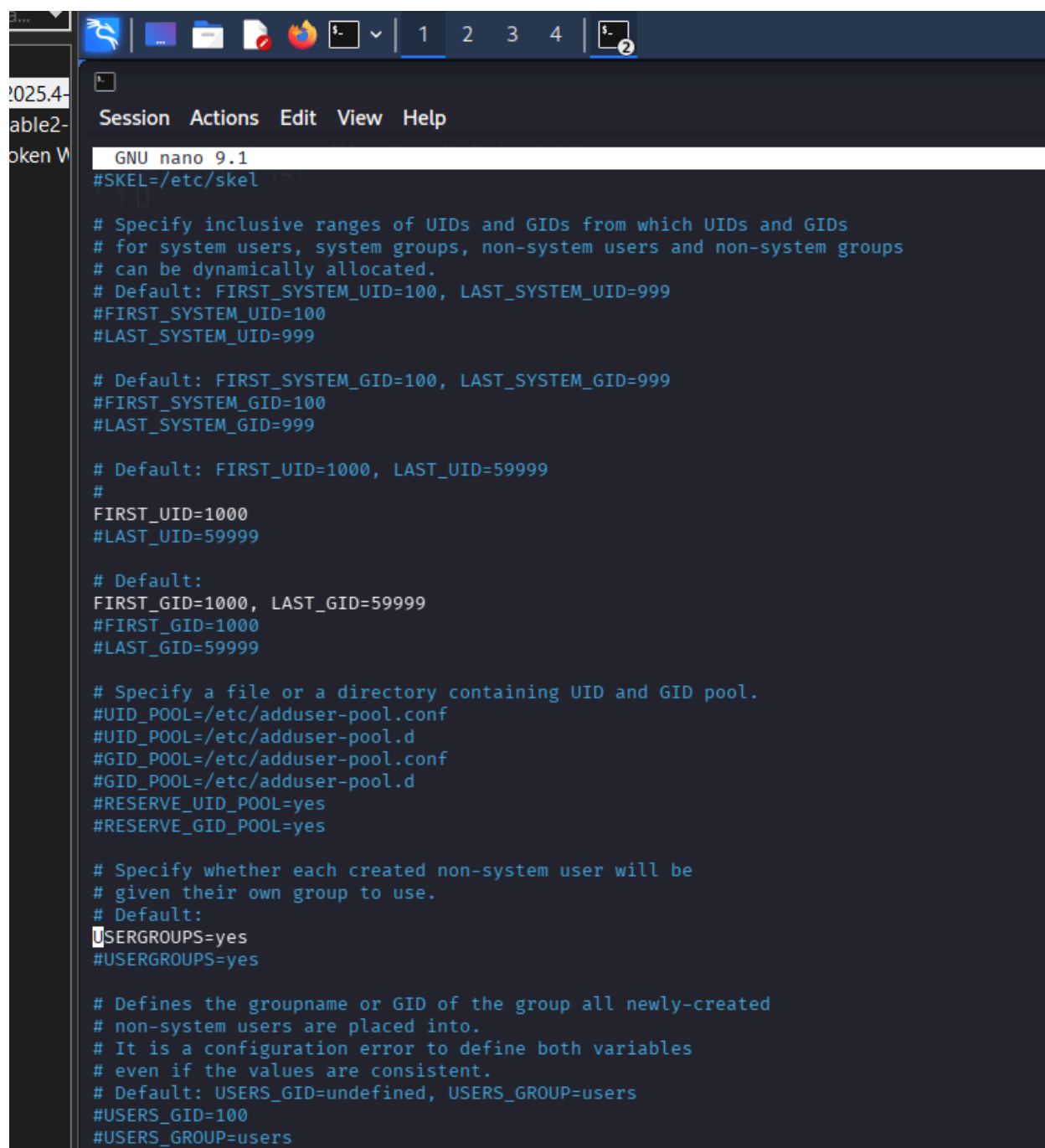
# The directory from which skeletal user configuration files
# will be copied.
# Default: SKEL=/etc/skel
#SKEL=/etc/skel

# Specify inclusive ranges of UIDs and GIDs from which UIDs and GIDs
# for system users, system groups, non-system users and non-system groups
# can be dynamically allocated.
# Default: FIRST_SYSTEM_UID=100, LAST_SYSTEM_UID=999
#FIRST_SYSTEM_UID=100
#LAST_SYSTEM_UID=999

# Default: FIRST_SYSTEM_GID=100, LAST_SYSTEM_GID=999
#FIRST_SYSTEM_GID=100
#LAST_SYSTEM_GID=999

# Default: FIRST_UID=1000, LAST_UID=59999
#FIRST_UID=1000
#LAST_UID=59999

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E
```



The image shows a terminal window with the nano text editor open. The editor displays the configuration file /etc/adduser.conf. The window's title bar includes standard Linux window controls and a taskbar with icons for a file manager, terminal, and other applications. The nano editor's menu bar shows 'Session', 'Actions', 'Edit', 'View', and 'Help'. The file content is as follows:

```
GNU nano 9.1
#SKEL=/etc/skel

# Specify inclusive ranges of UIDs and GIDs from which UIDs and GIDs
# for system users, system groups, non-system users and non-system groups
# can be dynamically allocated.
# Default: FIRST_SYSTEM_UID=100, LAST_SYSTEM_UID=999
#FIRST_SYSTEM_UID=100
#LAST_SYSTEM_UID=999

# Default: FIRST_SYSTEM_GID=100, LAST_SYSTEM_GID=999
#FIRST_SYSTEM_GID=100
#LAST_SYSTEM_GID=999

# Default: FIRST_UID=1000, LAST_UID=59999
#
FIRST_UID=1000
#LAST_UID=59999

# Default:
FIRST_GID=1000, LAST_GID=59999
#FIRST_GID=1000
#LAST_GID=59999

# Specify a file or a directory containing UID and GID pool.
#UID_POOL=/etc/adduser-pool.conf
#UID_POOL=/etc/adduser-pool.d
#GID_POOL=/etc/adduser-pool.conf
#GID_POOL=/etc/adduser-pool.d
#RESERVE_UID_POOL=yes
#RESERVE_GID_POOL=yes

# Specify whether each created non-system user will be
# given their own group to use.
# Default:
USERGROUPS=yes
#USERGROUPS=yes

# Defines the groupname or GID of the group all newly-created
# non-system users are placed into.
# It is a configuration error to define both variables
# even if the values are consistent.
# Default: USERS_GID=undefined, USERS_GROUP=users
#USERS_GID=100
#USERS_GROUP=users
```

The configuration file was examined to identify key user management parameters. Some settings listed in the lab manual were not available in the current Kali Linux version, while others such as FIRST_UID, LAST_UID, FIRST_GID, LAST_GID, and USERGROUPS were successfully located.

Exercise 3: Modify the Configuration

1. Change Default Home Directory Location:
 - Modify the DHOME value to /mnt/users in the adduser.conf file. Save the file.
2. Customize UID and GID Ranges:
 - Change FIRST_UID to 2000 and LAST_UID to 2999.
 - Change FIRST_GID to 3000 and LAST_GID to 3999.
3. Disable User-Specific Groups:
 - Set USERGROUPS=no in the configuration file.
4. Add Custom Files to the Skel Directory:
 - Create a custom file in /etc/skel:

```
echo "Welcome to your new account!" | sudo tee /etc/skel/welcome.txt
```

5. Set a Different Default Shell:
 - Change DSHELL to /bin/sh. Save the file.
-

Exercise 4: Test the Changes

1. Create a New User:
 - Use the adduser command to create a new user:

```
sudo adduser testuser
```

2. Verify the Changes:
 - Check the home directory of the new user:

```
ls /mnt/users/testuser
```

- Confirm that the `welcome.txt` file exists in the home directory.
- Verify the UID, GID, and default shell for the new user:

```
grep testuser /etc/passwd
```

Step 1 : Explore login.defs

1. Check the existing password, UID, and GID settings

```
grep -E  
'^(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE|UID_MIN|UID_MAX|GID_MIN|GID_MAX)' /etc/login.defs
```

2. Edit the login.defs configuration

```
sudo nano /etc/login.defs
```

Configured:

FROM:

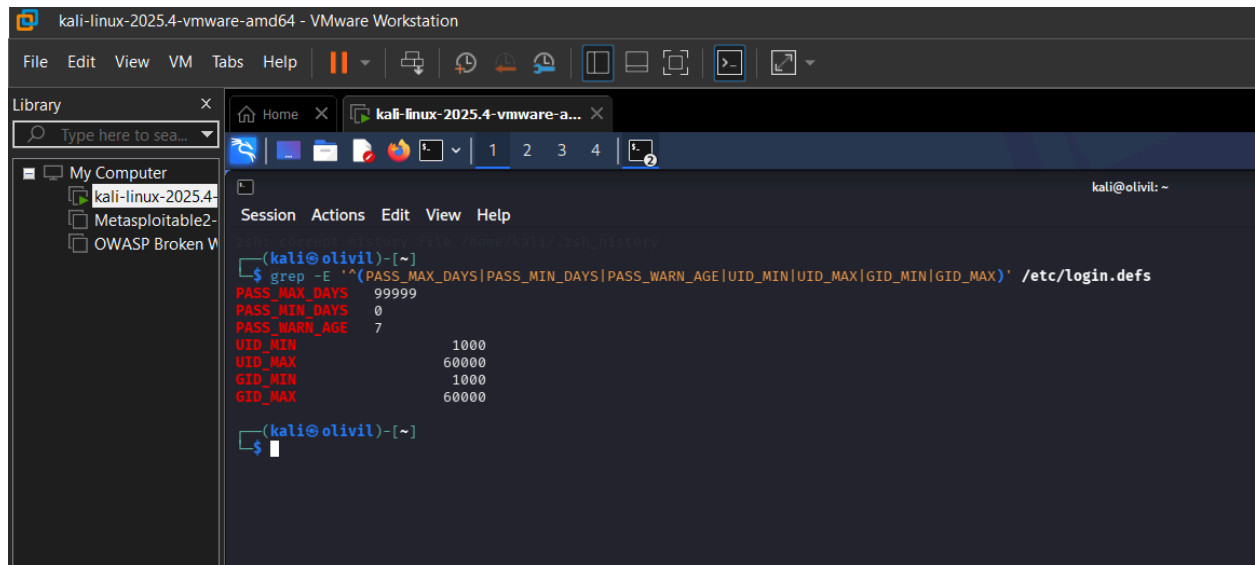
```
PASS_MAX_DAYS 99999  
PASS_MIN_DAYS 0  
PASS_WARN_AGE 7  
UID_MIN 1000  
UID_MAX 60000  
GID_MIN 1000  
GID_MAX 60000
```

TO:

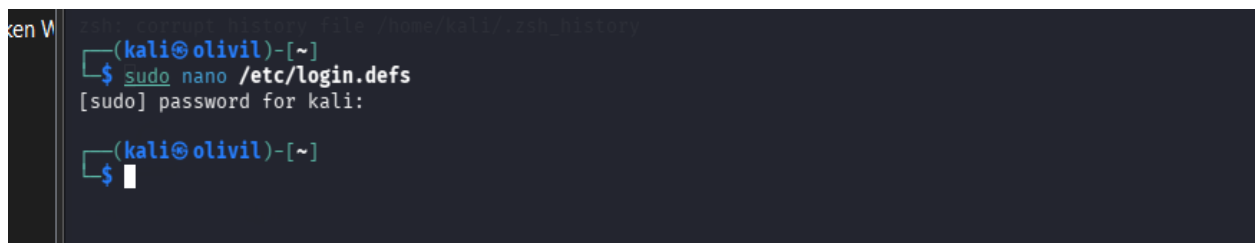
```
PASS_MAX_DAYS 90  
PASS_MIN_DAYS 7  
PASS_WARN_AGE 14  
UID_MIN 1000  
UID_MAX 60000  
GID_MIN 1000  
GID_MAX 60000
```

CTRL + O = ENTER

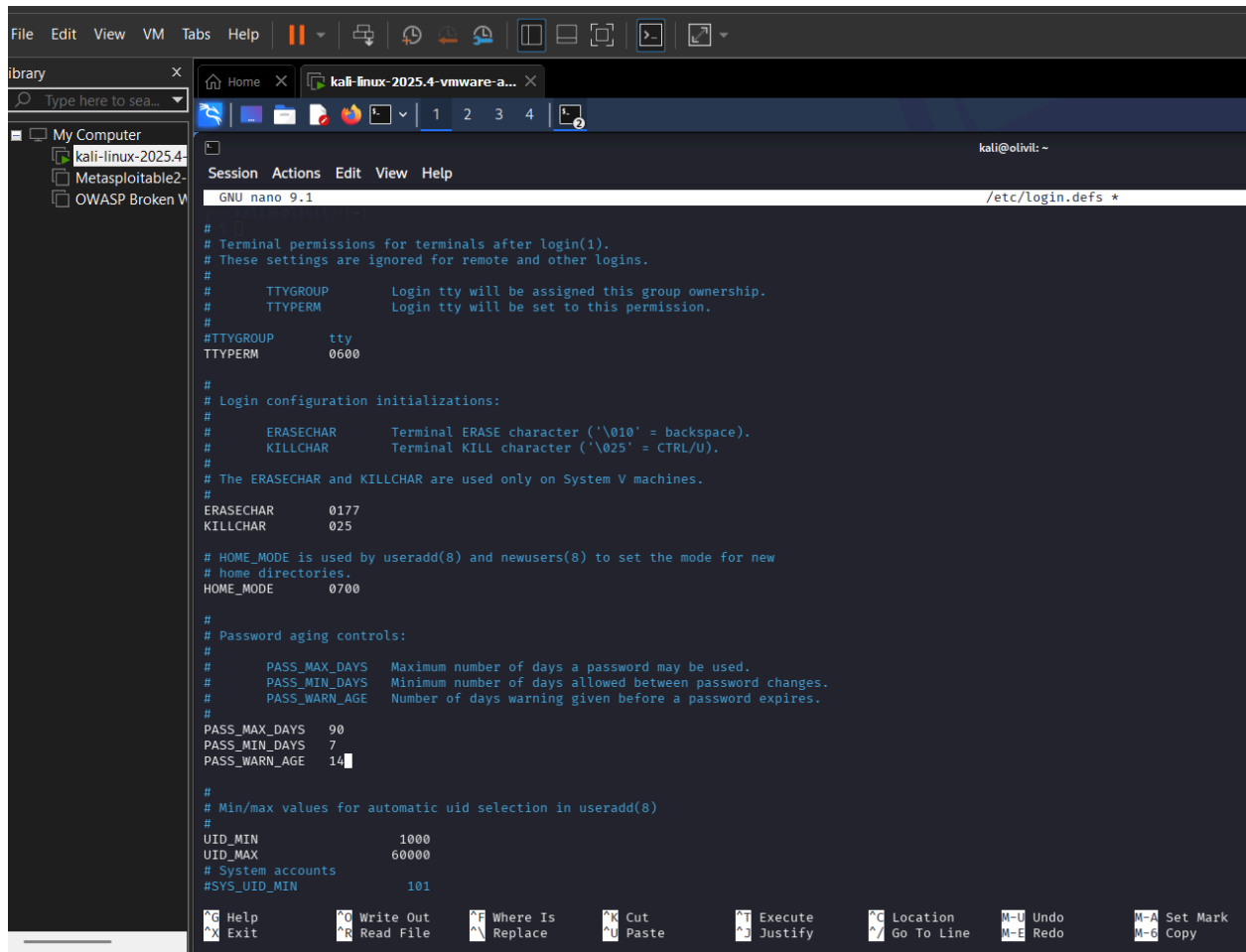
CTRL + X = EXIT FILE



```
kali-linux-2025.4-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
kali-linux-2025.4
Metasploitable2
OWASP Broken W
kali@olivil: ~
Session Actions Edit View Help
(kali@olivil)-[~]
$ grep -E '^(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE|UID_MIN|UID_MAX|GID_MIN|GID_MAX)' /etc/login.defs
PASS_MAX_DAYS 99999
PASS_MIN_DAYS 0
PASS_WARN_AGE 7
UID_MIN 1000
UID_MAX 60000
GID_MIN 1000
GID_MAX 60000
(kali@olivil)-[~]
$
```



```
ken W
(kali@olivil)-[~]
$ sudo nano /etc/login.defs
[sudo] password for kali:
(kali@olivil)-[~]
$
```

A screenshot of a Kali Linux virtual machine environment. The main window shows the nano text editor editing the /etc/login.defs file. The file contains various system configuration settings for login and password management. The settings for password aging (PASS_MAX_DAYS, PASS_MIN_DAYS, PASS_WARN_AGE) are highlighted in blue. The existing UID and GID ranges are also visible. The terminal prompt is 'kali@olivil: ~'.

```
File Edit View VM Tabs Help
library
Type here to search
My Computer
kali-linux-2025.4
Metasploitable2
OWASP Broken V
kali@olivil: ~
Session Actions Edit View Help
GNU nano 9.1 /etc/login.defs *
#
# Terminal permissions for terminals after login(1).
# These settings are ignored for remote and other logins.
#
#       TTYGROUP      Login tty will be assigned this group ownership.
#       TTYPERM       Login tty will be set to this permission.
#
#TTYGROUP      tty
#TTYPERM       0600
#
# Login configuration initializations:
#
#       ERASECHAR      Terminal ERASE character ('\010' = backspace).
#       KILLCHAR       Terminal KILL character ('\025' = CTRL/U).
#
# The ERASECHAR and KILLCHAR are used only on System V machines.
#
#ERASECHAR     0177
#KILLCHAR      025
#
# HOME_MODE is used by useradd(8) and newusers(8) to set the mode for new
# home directories.
#HOME_MODE     0700
#
# Password aging controls:
#
#       PASS_MAX_DAYS  Maximum number of days a password may be used.
#       PASS_MIN_DAYS  Minimum number of days allowed between password changes.
#       PASS_WARN_AGE  Number of days warning given before a password expires.
#
#PASS_MAX_DAYS 90
#PASS_MIN_DAYS 7
#PASS_WARN_AGE 14
#
# Min/max values for automatic uid selection in useradd(8)
#
#UID_MIN       1000
#UID_MAX       60000
# System accounts
#SYS_UID_MIN   101
#
# Help      Write Out  Where Is  Cut      Execute  Location  M-U Undo  M-A Set Mark
# Exit     Read File  Replace  Paste    Justify  Go To Line M-E Redo  M-C Copy
```

I first reviewed the existing login.defs settings to establish the current account-security configuration. I then opened the file and updated the password policy to require a maximum password age of 90 days, a minimum age of 7 days, and a 14-day warning period. The existing UID and GID ranges already matched the required security policy, so they were left unchanged.

Step 2: Tested User and Applied Password Policy

Commands used:

1. Verify the updated login.defs settings

```
grep -E  
'^(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE|UID_MIN|UID_MAX|GID_MIN|GID_MAX)' /etc/login.defs
```

2. Attempt to create the test user

```
sudo useradd testuser (which user already exist from previous labs)
```

3. Verified the existing testuser account

```
id testuser
```

4. Checked the user's original password-aging settings

```
sudo chage -l testuser
```

5. Applied the required password policy

```
sudo chage -m 7 -M 90 -W 14 testuser
```

6. Verified the updated password-aging settings

```
sudo chage -l testuser
```

```
(kali@olivil)-[~]
$ grep -E '^(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE|UID_MIN|UID_MAX|GID_MIN|GID_MAX)' /etc/login.defs
PASS_MAX_DAYS    90
PASS_MIN_DAYS     7
PASS_WARN_AGE    14
UID_MIN           1000
UID_MAX          60000
GID_MIN           1000
GID_MAX          60000

(kali@olivil)-[~]
$ sudo useradd testuser
useradd: user 'testuser' already exists

(kali@olivil)-[~]
$ id testuser
uid=1001(testuser) gid=1001(testuser) groups=1001(testuser),100(users)

(kali@olivil)-[~]
$ sudo chage -l testuser
Last password change                : May 11, 2026
Password expires                    : Feb 23, 2300
Password inactive                    : never
Account expires                     : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7

(kali@olivil)-[~]
$ sudo chage -m 7 -M 90 -W 14 testuser

(kali@olivil)-[~]
$ sudo chage -l testuser
Last password change                : May 11, 2026
Password expires                    : Aug 09, 2026
Password inactive                    : never
Account expires                     : never
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14

(kali@olivil)-[~]
$
```

verified that the required login.defs settings were correctly configured. The testuser account already existed, so instead of creating a duplicate account, I checked its existing settings and applied the required password policy using **chage**. The final verification confirmed a 7-day minimum password age, 90-day maximum password age, and 14-day warning period.

Step 3: Set and Test the testuser Password

Commands used:

1. Set a password for testuser

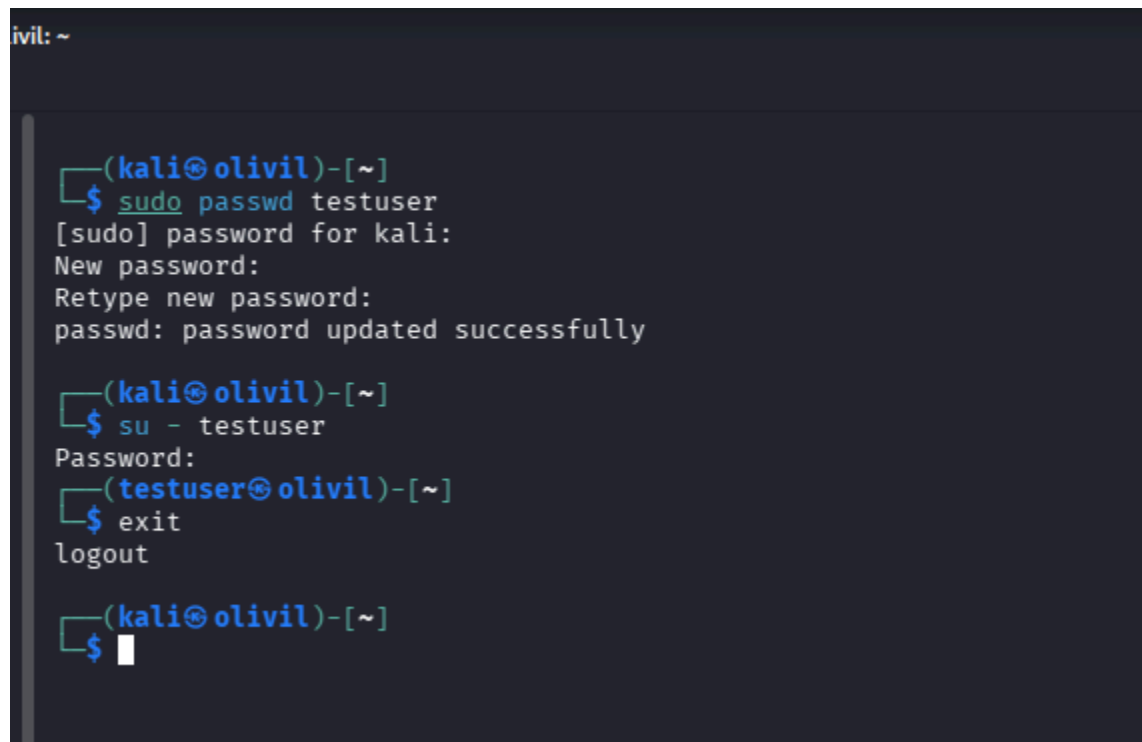
`sudo passwd testuser`

2. Switch to the testuser account

`su - testuser`

3. Exit the testuser session

`exit`



```
ivilt: ~  
  
(kali@olivil)-[~]  
$ sudo passwd testuser  
[sudo] password for kali:  
New password:  
Retype new password:  
passwd: password updated successfully  
  
(kali@olivil)-[~]  
$ su - testuser  
Password:  
(testuser@olivil)-[~]  
$ exit  
logout  
  
(kali@olivil)-[~]  
$
```

successfully set a password for the testuser account and verified that the account could be accessed by switching to it with `su - testuser`. After confirming successful login, I exited the test account and returned to the normal kali user session.

Lab 4: Advanced User Access Control and Security Auditing

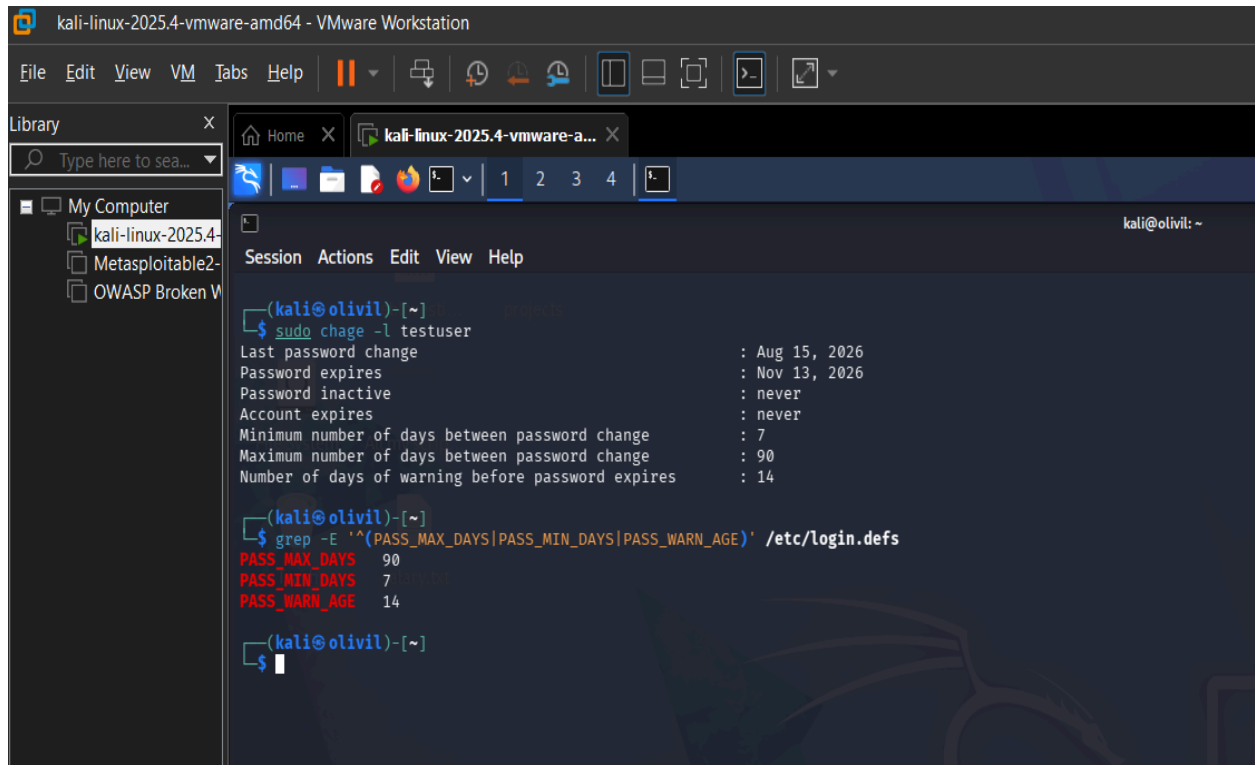
Part 1: Password Policy Configuration

Exercise 1.1: configure Password Aging

Commands used:

```
sudo chage -l testuser
```

```
grep -E '^(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE)'  
/etc/login.defs
```



```
(kali@olivil)-[~]
$ sudo chage -l testuser
Last password change           : Aug 15, 2026
Password expires                : Nov 13, 2026
Password inactive              : never
Account expires                : never
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14

(kali@olivil)-[~]
$ grep -E '(PASS_MAX_DAYS|PASS_MIN_DAYS|PASS_WARN_AGE)' /etc/login.defs
PASS_MAX_DAYS 90
PASS_MIN_DAYS 7
PASS_WARN_AGE 14

(kali@olivil)-[~]
$
```

testuser was verified to have a 7-day minimum password age, 90-day maximum password age, and 14-day warning period. The system-wide /etc/login.defs settings matched these required values.

Question 1.1: Why is it important to set a minimum number of days between password changes?

A minimum password age prevents users from repeatedly changing their password in a short period just to bypass password-history rules and return to an old password. It helps ensure that passwords are kept for a reasonable period and improves account security.

Question 1.2: What are the security implications of setting PASS_MAX_DAYS too high or too low?

- Too high: A compromised or weak password may remain valid for a long time, increasing the risk of unauthorized access.
 - Too low: Users are forced to change passwords too frequently, which can lead to password fatigue and encourage users to choose weaker or predictable passwords.
-

Exercise 1.2: Implement Password Complexity Requirements

Commands used

```
sudo apt update
sudo apt install libpam-pwquality -y
sudo nano /etc/security/pwquality.conf
```

Added to `pwquality.conf`:

```
minlen = 12
dcredit = -1
ucredit = -1
lcredit = -1
ocredit = -1
difok = 3
maxrepeat = 2
usercheck = 1
```

Tested the password policy with:

```
sudo passwd testuser
```

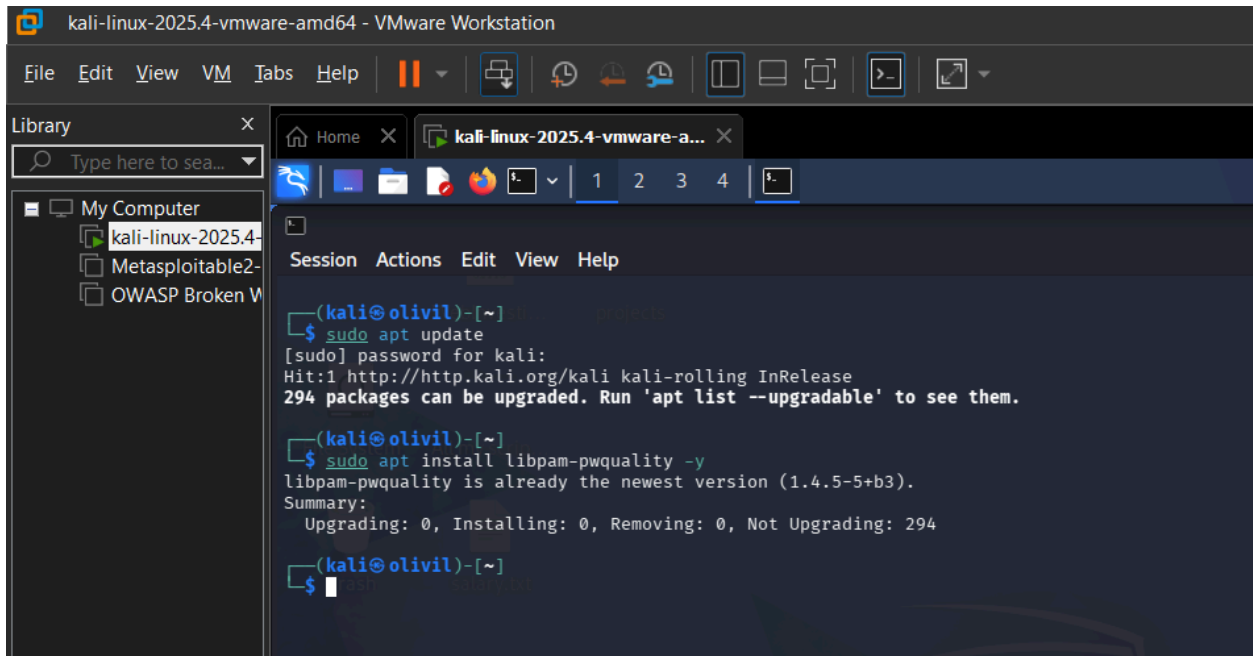
Weak password tested:

```
password
```

It was rejected with a dictionary-check warning.

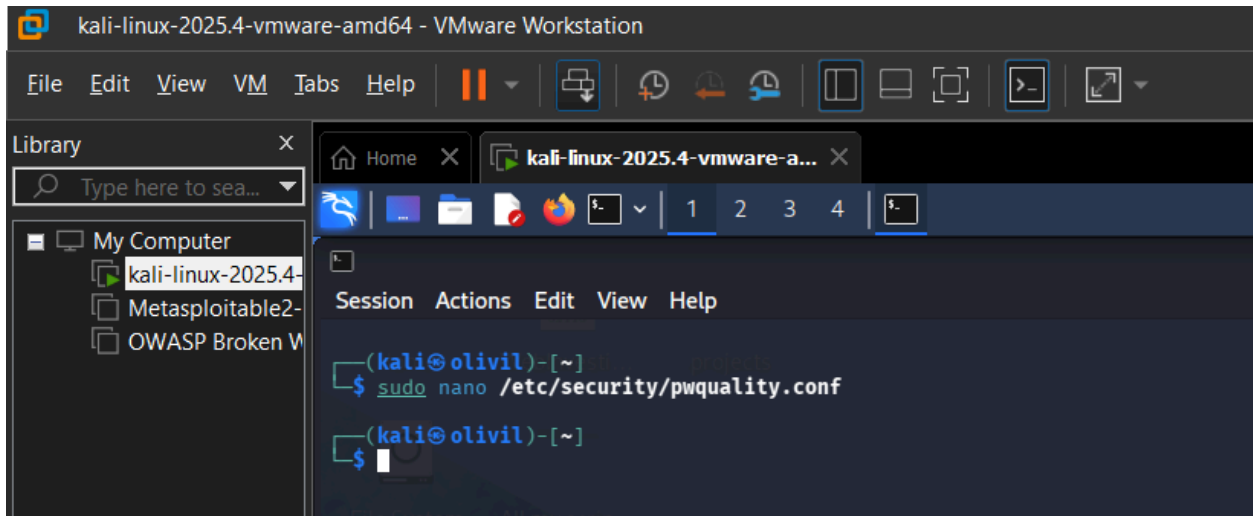
A compliant password was then tested:

BlueSky@2026!



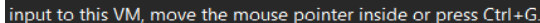
The screenshot shows a VMware Workstation window titled "kali-linux-2025.4-vmware-amd64 - VMware Workstation". The interface includes a menu bar (File, Edit, View, VM, Tabs, Help), a toolbar with icons for file operations and VM control, and a sidebar with a "Library" pane showing "My Computer" with folders for "kali-linux-2025.4-", "Metasploitable2-", and "OWASP Broken V...". The main terminal window displays the following commands and output:

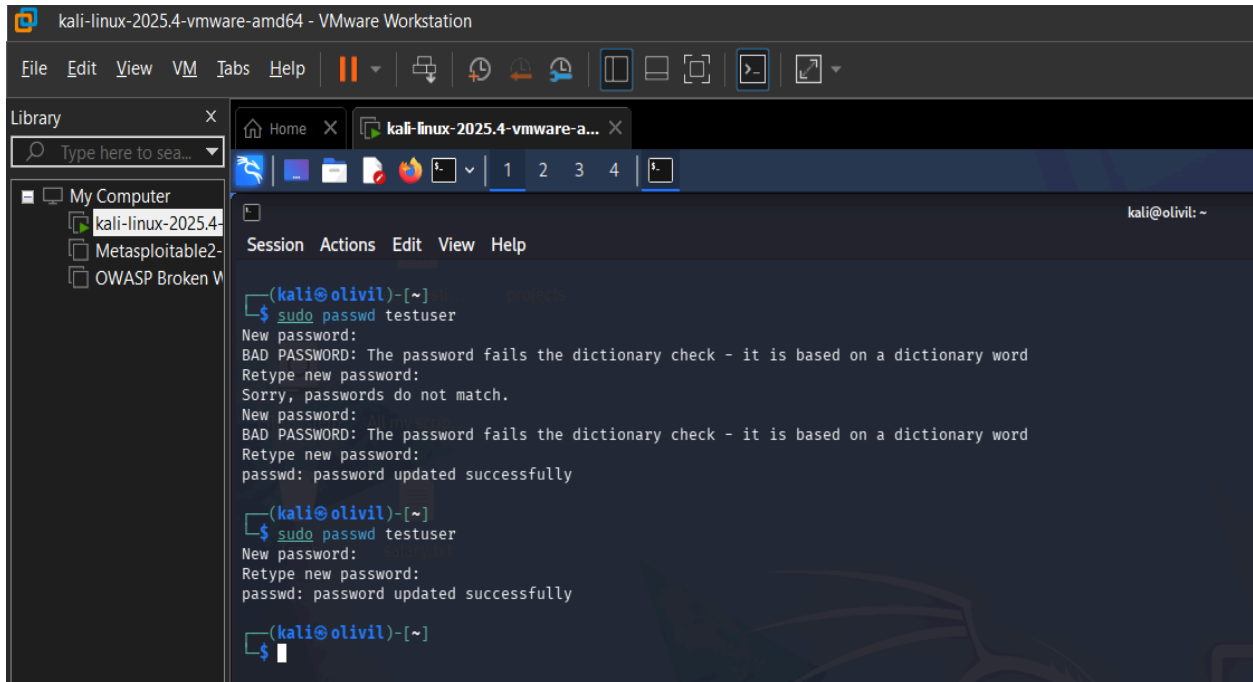
```
(kali@olivil)-[~]  
$ sudo apt update  
[sudo] password for kali:  
Hit:1 http://http.kali.org/kali kali-rolling InRelease  
294 packages can be upgraded. Run 'apt list --upgradable' to see them.  
  
(kali@olivil)-[~]  
$ sudo apt install libpam-pwquality -y  
libpam-pwquality is already the newest version (1.4.5-5+b3).  
Summary:  
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 294  
  
(kali@olivil)-[~]  
$
```



The screenshot shows the same VMware Workstation window. The terminal window displays the following commands and output:

```
(kali@olivil)-[~]  
$ sudo nano /etc/security/pwquality.conf  
  
(kali@olivil)-[~]  
$
```



A screenshot of a Kali Linux virtual machine running in VMware Workstation. The terminal window shows a user attempting to set a password for the 'testuser' using the 'passwd' command. The first attempt is rejected because the password is a dictionary word. The second attempt is also rejected for the same reason. The third attempt is successful. The terminal output is as follows:

```
(kali@olivil)-[~]  
$ sudo passwd testuser  
New password:  
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word  
Retype new password:  
Sorry, passwords do not match.  
New password:  
BAD PASSWORD: The password fails the dictionary check - it is based on a dictionary word  
Retype new password:  
passwd: password updated successfully  
  
(kali@olivil)-[~]  
$ sudo passwd testuser  
New password:  
Retype new password:  
passwd: password updated successfully  
  
(kali@olivil)-[~]  
$
```

I configured the password complexity requirements and tested the policy with both weak and strong passwords. The weak password was rejected, while the compliant password was accepted successfully.

Question 1.3: What happens when you try to set a password that doesn't meet the complexity requirements?

When a password does not meet the complexity requirements, the system rejects it and displays an error explaining that the password is too weak or fails a required check.

Question 1.4: How do password complexity requirements help prevent brute-force attacks?

Password complexity makes passwords harder to guess by requiring a combination of different character types and sufficient length. This increases the number of possible password combinations, making brute-force attacks more difficult.